

Assignment-2.4

“Connecting and Testing Peripherals”

TASK-1 Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo.

1. **Connect Hardware:** Plug the USB webcam directly into a spare USB port on your PC/laptop.
2. **Driver Check:** Windows 10/11 will automatically initialize generic UVC drivers. If your webcam has custom software (e.g., Logitech G HUB), install it from the manufacturer's site.
3. **Open Camera App:** search for **Camera**, and open the built-in Windows Camera app.
4. **Capture Test Photo:** Verify that the camera stream appears, then click the **Photo** button to capture an image.
 - **Verification:** Navigate to **File Explorer > Pictures > Camera Roll** to confirm the captured photo file is saved and readable.

TASK-2 Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

1. **Physical Setup:** Connect the printer to the PC via USB (or connect it to the local network). Power it on and load paper and ink/toner.
2. **Add Device:** Navigate to **Settings > Devices > Printers & scanners** (or *Bluetooth & devices > Printers & scanners*). Click **Add device**.
3. **Driver Installation:** If Windows does not auto-detect the driver, download the driver installation package directly from the manufacturer's website (HP, Epson, Canon, etc.) and run the wizard.
4. **Print Test Page:** Under *Printers & scanners*, select your printer, click **Manage**, and click **Print a test page**.
 - **Verification:** Confirm that the printer physically prints out the Windows Test Page showing driver information and alignment graphics.

TASK-3 Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem. Check for error icons and use the 'Scan for hardware changes' option after reconnecting.

1. Simulate the Disconnection:

- Unplug the printer's USB cable from your computer while the PC is turned on.
- *What happens:* Windows will show the printer as "Offline" or remove it from your active devices list.

2. Open Device Manager:

- Right-click the Windows **Start Button** and click **Device Manager** (or press **Win + X** and select **Device Manager**).

3. Check the Status & Plug It Back In:

- Look under **Print queues** or **Universal Serial Bus controllers** in Device Manager.
- If the printer is disconnected, its icon might disappear or show a greyed-out icon / yellow error triangle.
- Plug the printer's USB cable back in firmly.

4. Scan for Hardware Changes:

- Click **Action** on the top menu bar of Device Manager.
- Select **Scan for hardware changes**.
- *What happens:* This forces Windows to check all USB ports right away and recognize the printer.

TASK-4 A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications. Check if another app is using the webcam or if Zoom permissions are enabled.

1. **Close Conflicting Applications:** Windows allows only one application to access a webcam stream at a time. Close the Windows Camera app, Skype, or browser tabs that might be holding an active lock on the camera sensor.
 2. **Check Desktop App Permissions:** Go to **Settings > Privacy > Camera**. Ensure "*Allow apps to access your camera*" is **ON**, and verify that "*Allow desktop apps to access your camera*" is enabled (this controls Zoom access).
 3. **Select Camera in Zoom Settings:** Open Zoom, click your profile icon / gear icon (**Settings**), go to **Video**, and under the **Camera** dropdown menu, select your physical USB webcam instead of a virtual camera output (e.g., OBS Virtual Camera).
- **Verification:** Start a Zoom test meeting and confirm that the video preview loads properly without throwing a "*Cannot start video*" error.